



CAVITE

Company Attractiveness & Talent Intelligence Viability
Engine

A Hybrid Probabilistic–Neural System for Startup Hiring–Quality
Prediction

Team CAVITE

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Phase 1 (Advanced ML) + Phase 2 (Deep
Learning) — Undergraduate Project
Report

Problem & Why It Matters

The Problem

Predicting whether a startup is a good place to work is genuinely hard. Hiring desirability is **noisy, multi-dimensional**, and depends on funding stage, culture scores, growth trajectory, employee ratings, and the actual roles being hired for. No single signal is sufficient — and most signals are incomplete.

Why This Project

Sparse & Noisy Data

Indian startups lack standardised reporting. Reviews are sparse, funding data is patchy, and class boundaries are blurry by nature.

Better Signals Needed

Job seekers deserve more than raw Glassdoor averages. ML can surface latent patterns across dozens of features simultaneously.

Classical + Deep Learning Together

No prior work fuses structured startup signals with free-text role descriptions for hiring-quality classification at this scale.



Dataset Overview & Target Leakage Fix

Raw Dataset — 1,000 Indian Startups, 21 Features

Category	Features
Structural	Founded year, employee count, funding stage, total funding USD
Growth	Employee growth %, active job openings, AI roles focus
Sentiment	WLB, culture, growth opp, salary ratings (1–5)
Context	Sector, HQ city, remote-friendly, layoff flag
Text	Top hiring roles (semicolon-separated)

Three target classes: **Emerging**, **Growing**, **High Desirability**

⚠ Target Leakage Problem

The original label was generated by **thresholding a single column** — overall employee rating. A depth-3 decision tree on that one column achieved **CV Macro-F1 = 0.985**, making the task scientifically meaningless.

The Fix

We rebuilt labels via an **11-feature composite viability score** with domain-motivated weights and calibrated Gaussian noise $\epsilon \sim N(0, 0.15^2)$. Labels are assigned by tertile split, yielding a **near-perfectly balanced dataset (334 / 333 / 333)**. After the fix, the same depth-3 tree drops to **F1 $\approx$ 0.65** — confirming the task is now genuinely hard.

Feature Engineering & Key EDA Findings

Engineered Features

Hiring Pressure

Openings / employees — captures scaling speed

Log Funding

$\log(1 + \text{funding})$ — reduces right skew (raw skewness = 1.81)

Funding Per Employee

Capital efficiency proxy for startup maturity

Startup Age

2025 - founded year — company lifecycle signal

Review Density

Reviews per employee — coverage quality indicator

Growth × AI Focus

$\text{Growth\%} \times (1 + \text{AI focus})$ — captures tech-forward momentum

Key EDA Findings

→ Rating Overlap Across Classes

All four rating columns show mild negative skew ($|\text{skew}| < 0.2$) and heavy overlap between classes — no clean linear boundary.

→ Extreme Rating Multicollinearity

Pearson $r > 0.98$ between all five rating columns — a direct artefact of the original data-generating process. PCA is essential before GMM to avoid singular covariance matrices.

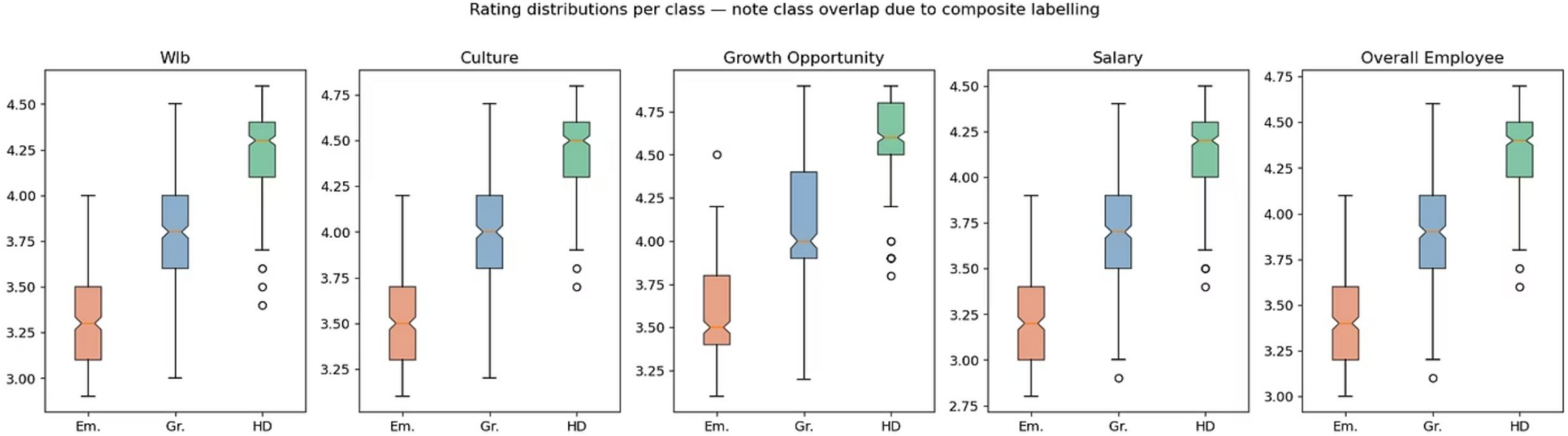
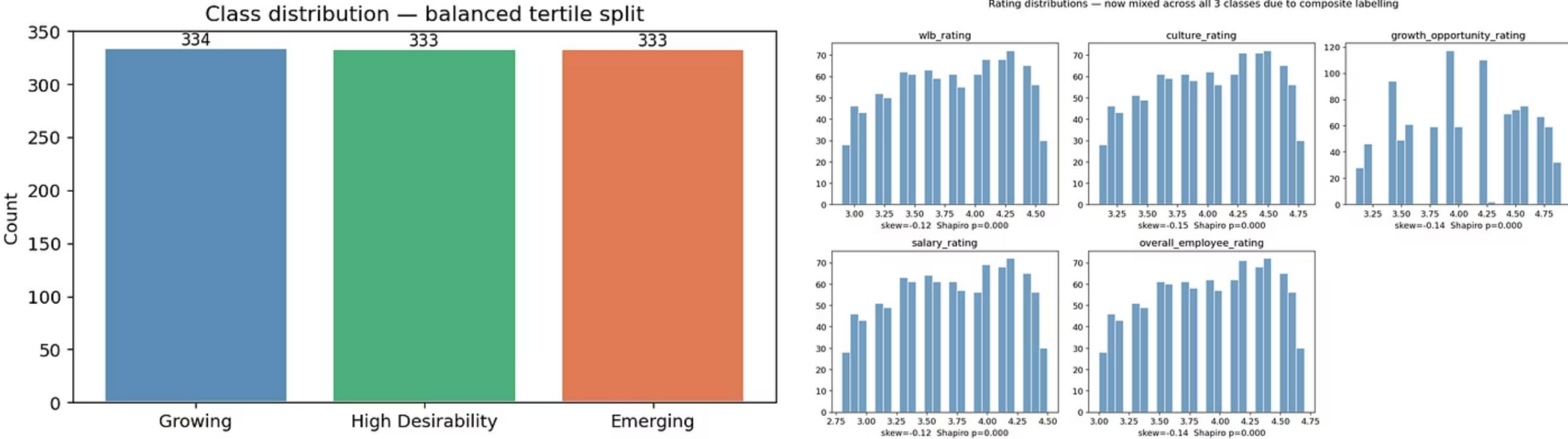
→ Heavy-Tailed Funding

Raw skewness = 1.81; $\log_{1p}$ transform reduces this substantially, validating the engineered log funding feature.

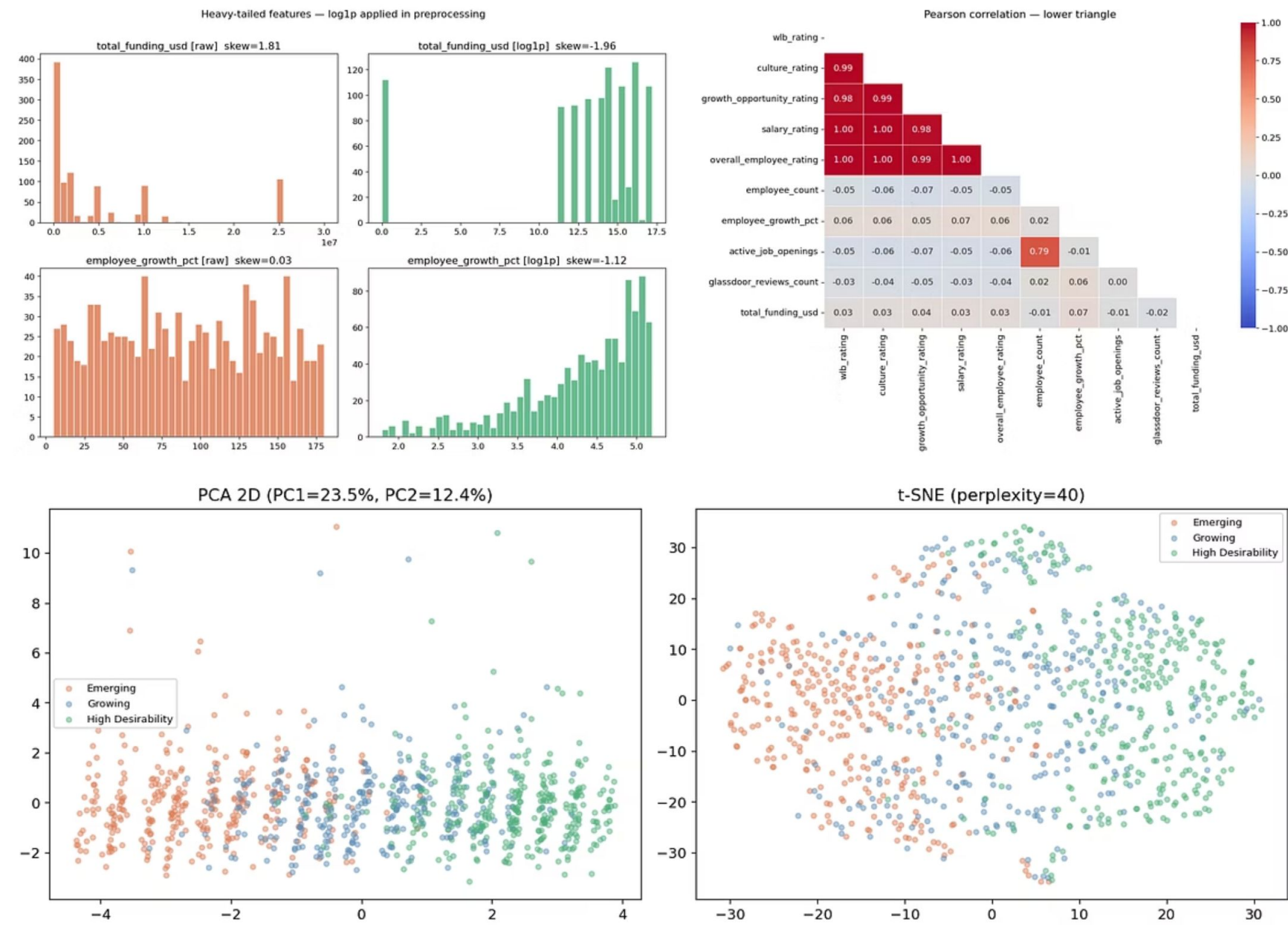
→ Non-Linear Class Overlap

PCA ($\text{PC1} = 23.5\%$, $\text{PC2} = 12.4\%$) and t-SNE both show heavy overlap with no visible separating boundary — directly motivating non-linear models.

Supporting EDA Visuals



Supporting EDA Visuals





PHASE 1

Advanced ML Models

Model A — Gaussian Mixture Model

Models data as a weighted mixture of K Gaussians, fit via EM. Chosen because its **soft posterior probabilities** (per-company cluster uncertainty) can be recycled as features in Phase 3. $K=3$ matches ground-truth classes; BIC model selection confirms full-covariance structure.

Model B — SVM with RBF Kernel

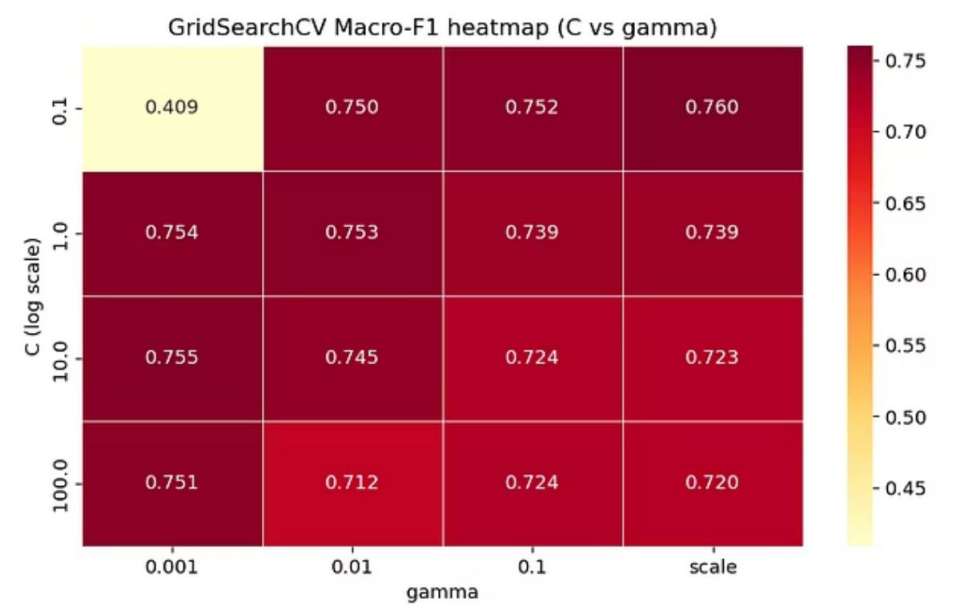
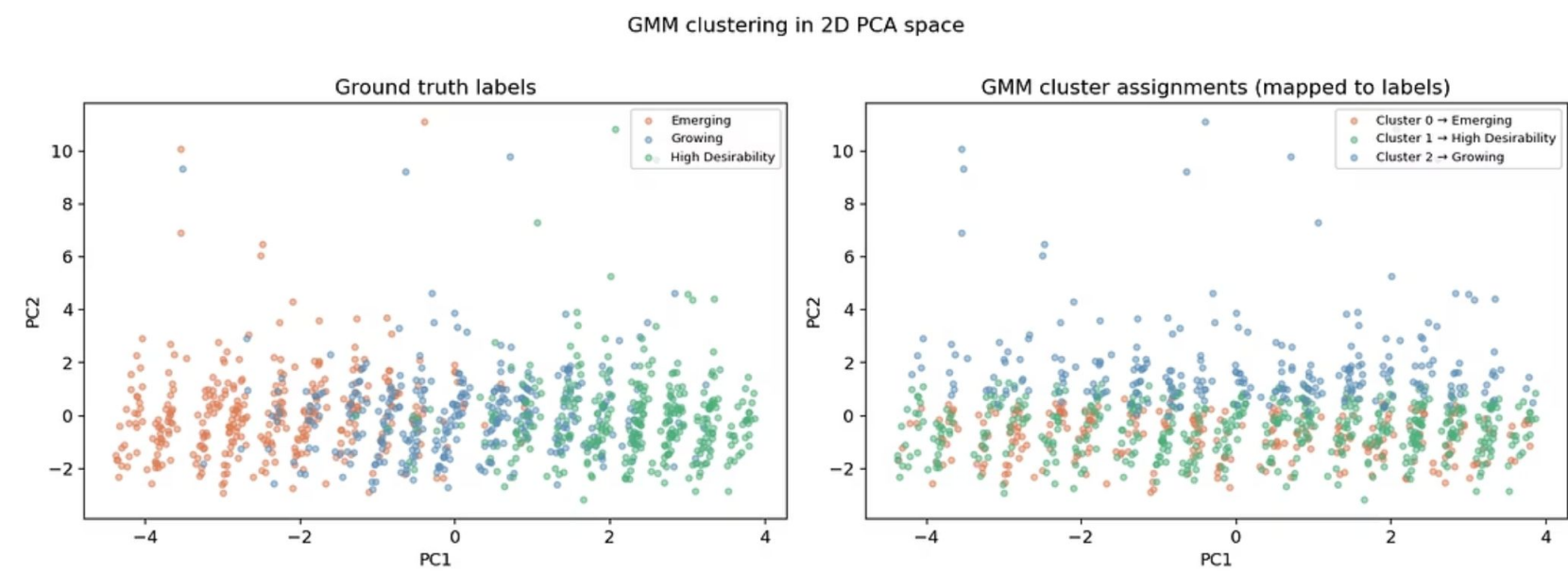
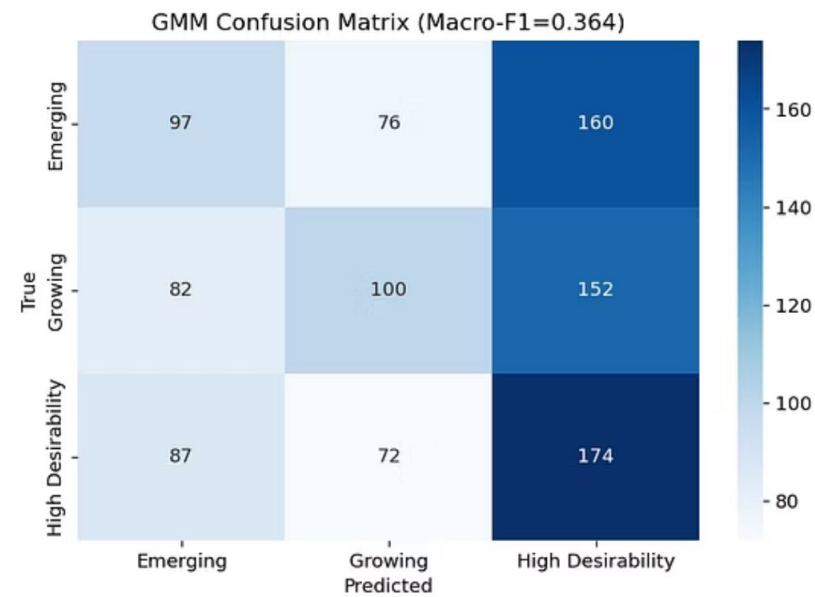
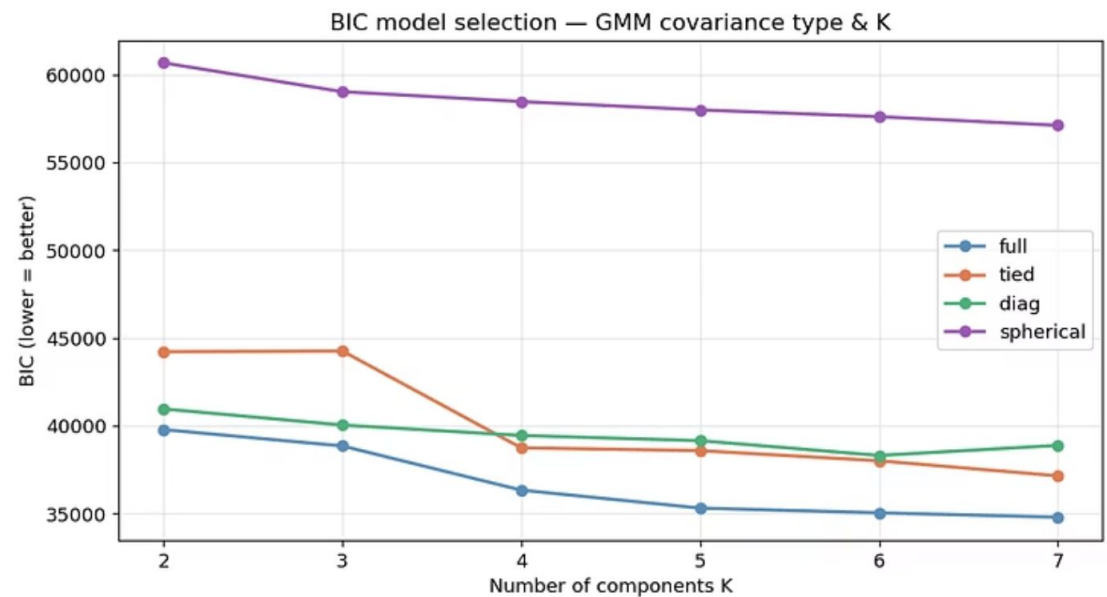
Finds a maximum-margin hyperplane in a kernel-induced Hilbert space. RBF kernel $K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2)$ satisfies Mercer's theorem. Chosen because EDA showed **non-linear class overlap** in PCA/t-SNE space, directly ruling out a linear kernel. Best: $C=0.1$, $\gamma=\text{scale}$.

Model C — XGBoost

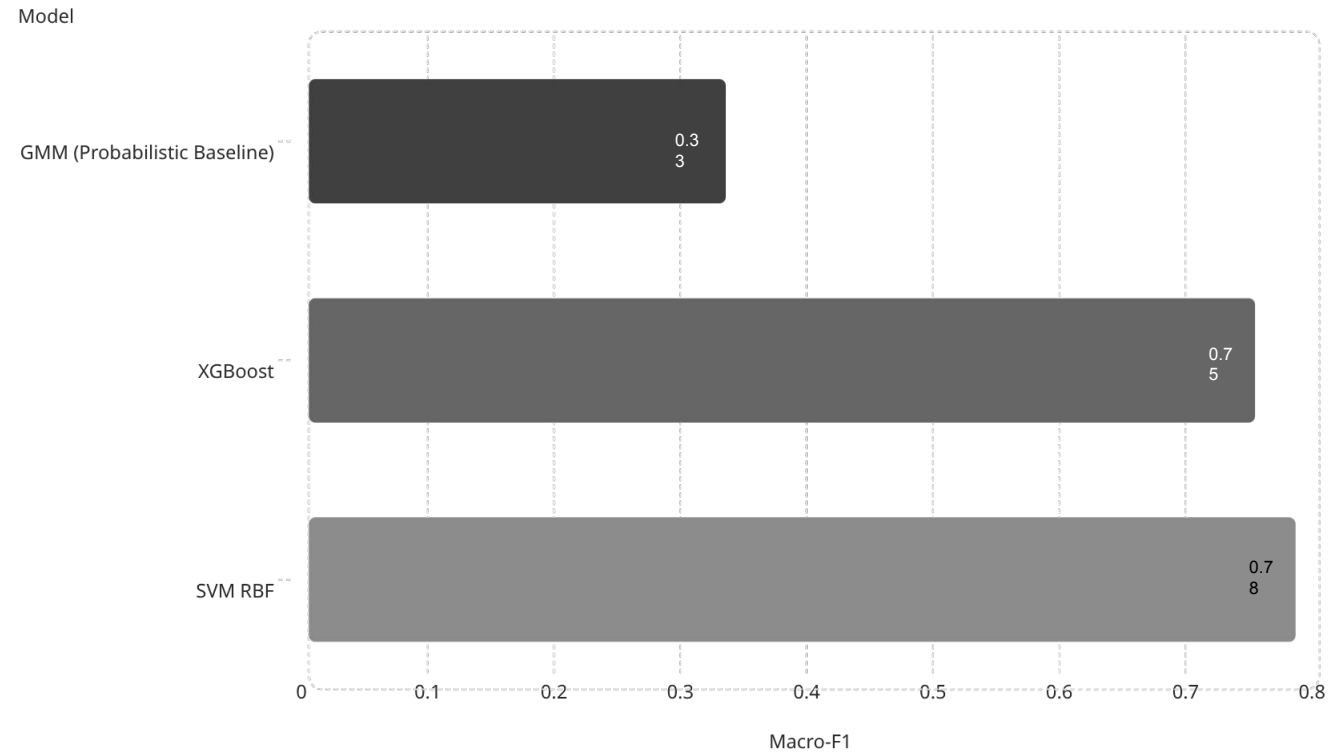
Builds trees sequentially, each correcting pseudo-residuals of the multi-class softmax log-loss. Regularisation term $\Omega(f) = \gamma T + \frac{1}{2} \lambda \|w\|^2$ prevents overfitting. Chosen for **explicit regularisation, SHAP interpretability**, and strong performance on moderately-sized tabular datasets. RandomizedSearchCV over 7 hyperparameters; early stopping on val log-loss.

 Text (top hiring roles) encoded with **TF-IDF** (30 bigram features, sublinear TF) for Phase 1. Phase 2 feeds raw text directly to DistilBERT.

Model Exploration Visuals



Model Performance Comparison



GMM — F1: 0.330

Unsupervised baseline. Cluster-to-label mapping concentrates errors on "High Desirability." Expected failure — class boundaries are close and overlapping. Soft posteriors retained as uncertainty features for Phase 3.

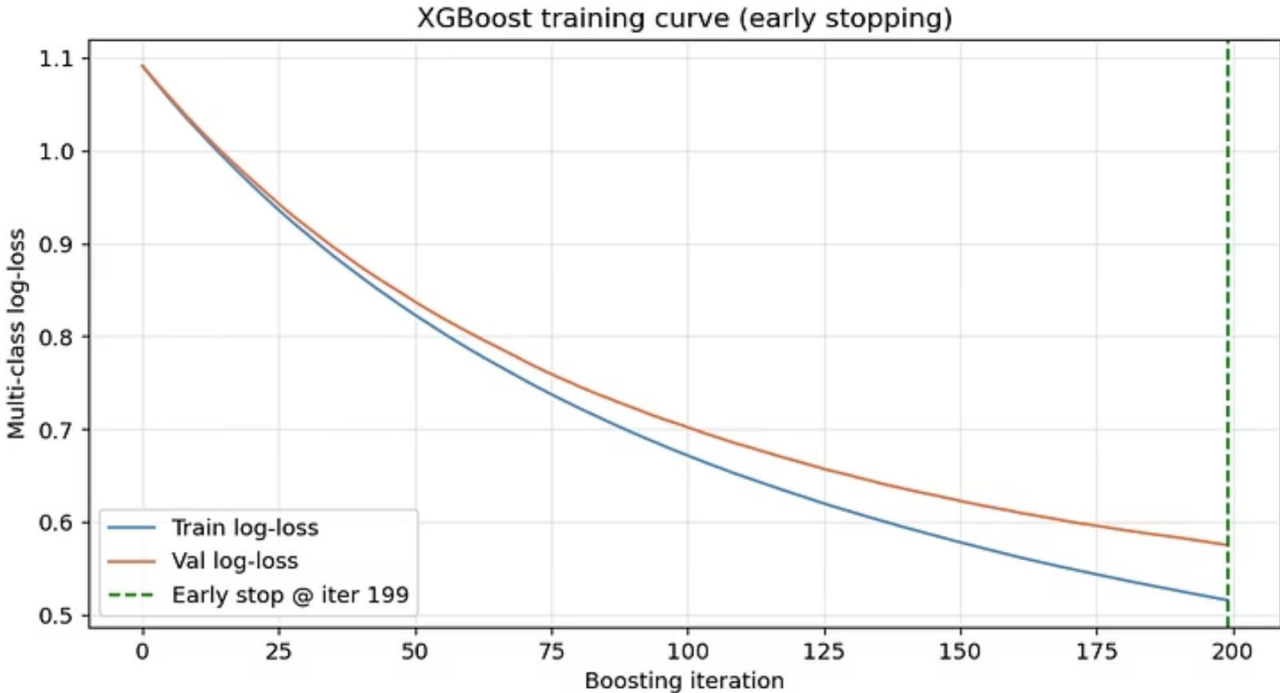
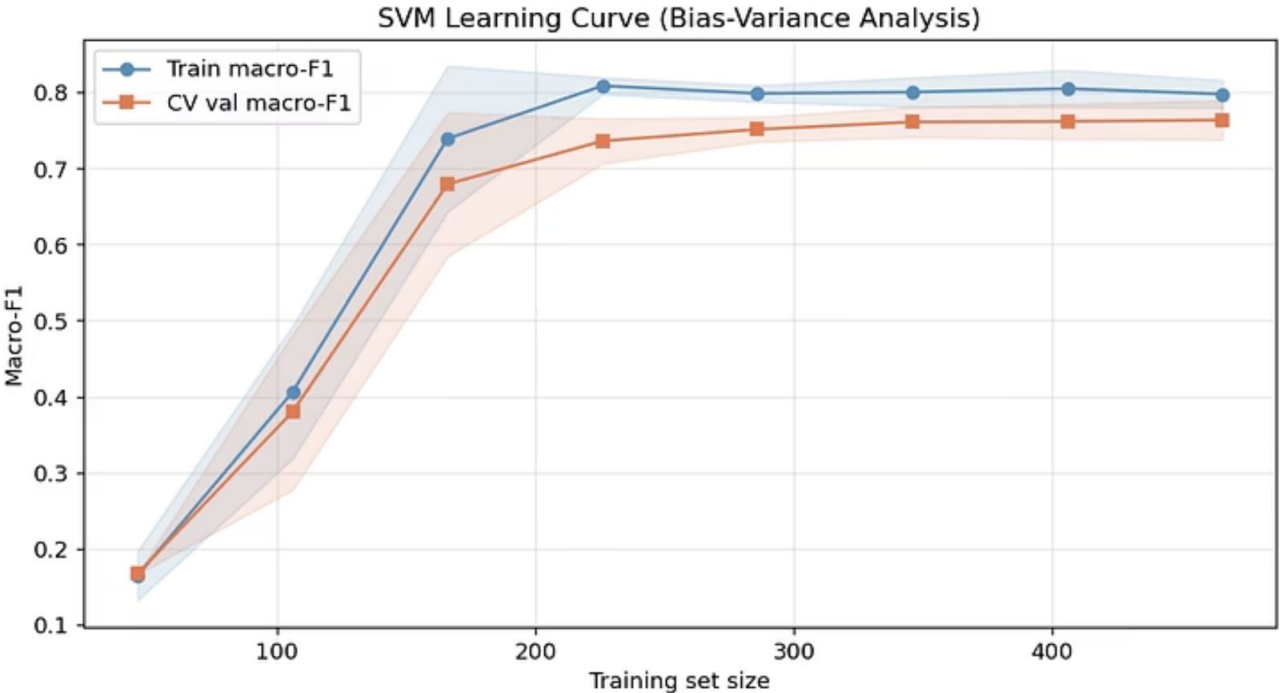
SVM RBF — F1: 0.781

Best Phase 1 model. **Zero Emerging ↔ HD confusions** — RBF correctly separates the most distant classes. Learning curves converge with train-val gap < 0.05, confirming no significant overfitting.

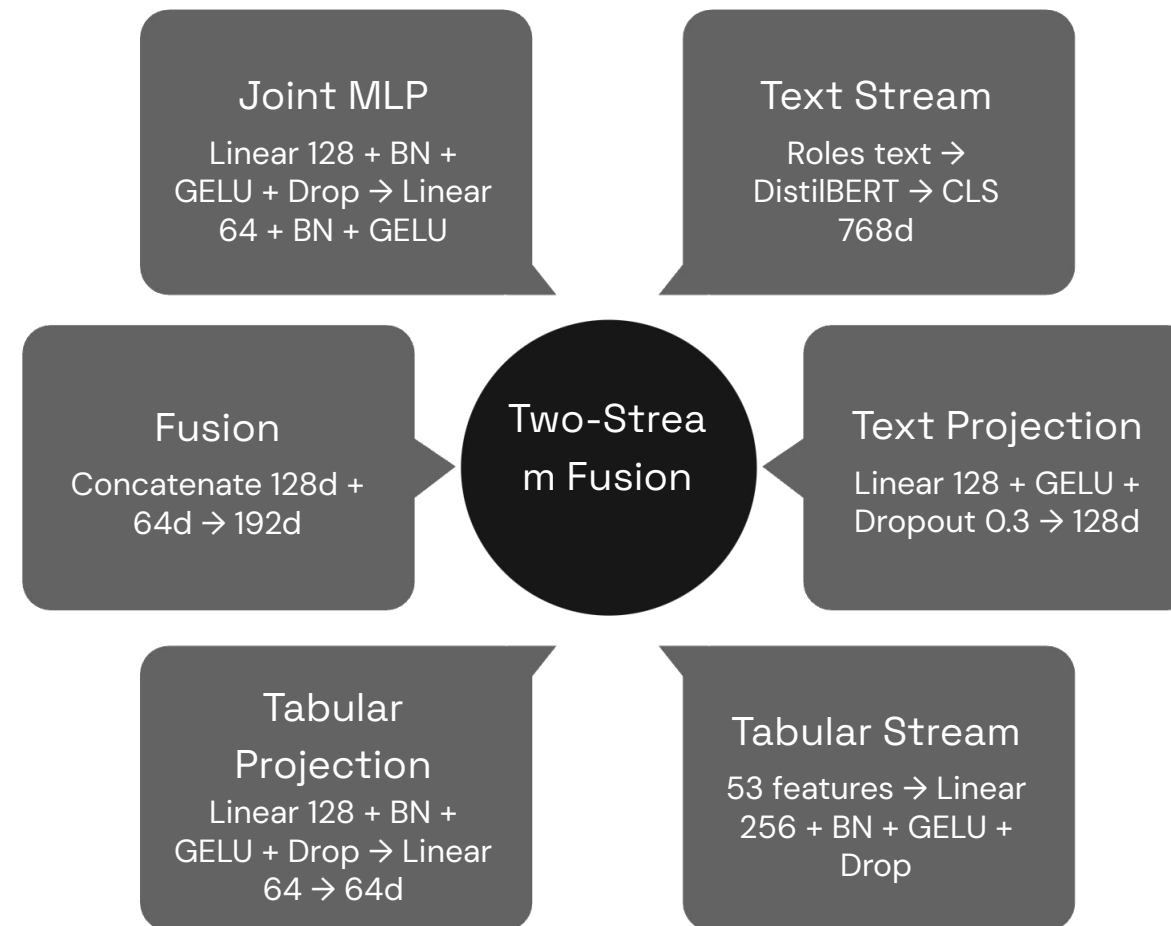
XGBoost — F1: 0.749

Close runner-up. Errors concentrate at adjacent-class boundaries (9 Emerging→Growing, 12 HD→Growing) — samples genuinely ambiguous due to the noise term ϵ in our composite score.

Training Curves



Deep Learning Hybrid Architecture



A single model cannot handle both text semantics and dense tabular interactions optimally. The two-stream fusion architecture processes each modality with the right inductive bias, then merges representations into a joint 192-dimensional space for final classification.

He Initialisation

$W \sim N(0, 2/n_{in})$ for GELU layers — keeps activation variance constant across all 4 MLP layers.

Batch Normalisation

Reduces internal covariate shift; enables LR = 10^{-3} for MLP head while DistilBERT fine-tunes at 2×10^{-5} .

Dropout $p=0.3$

Critical for preventing overfitting on 700 training samples. Scaled by $1/(1-p)$ at inference.

AdamW + Differential LR

Linear warmup (10% of steps) then decay to zero. DistilBERT: 2×10^{-5} ; randomly-initialised heads: 10^{-3} .

Training Results & Full Ablation Study

Hybrid Model Training

Val loss stabilises around epoch 5; val Macro-F1 peaks ≈ 0.80 before early stopping (patience = 3) saves the best checkpoint. The persistent train–val loss gap is expected: DistilBERT has 66M parameters and can perfectly memorise 700 training samples, but Dropout + BN + early stopping keep val performance strong. All errors are between **adjacent classes only** — zero Emerging HD confusion.

0.783

DL Hybrid F1

Best result across all 6 configurations

0.781

SVM RBF F1

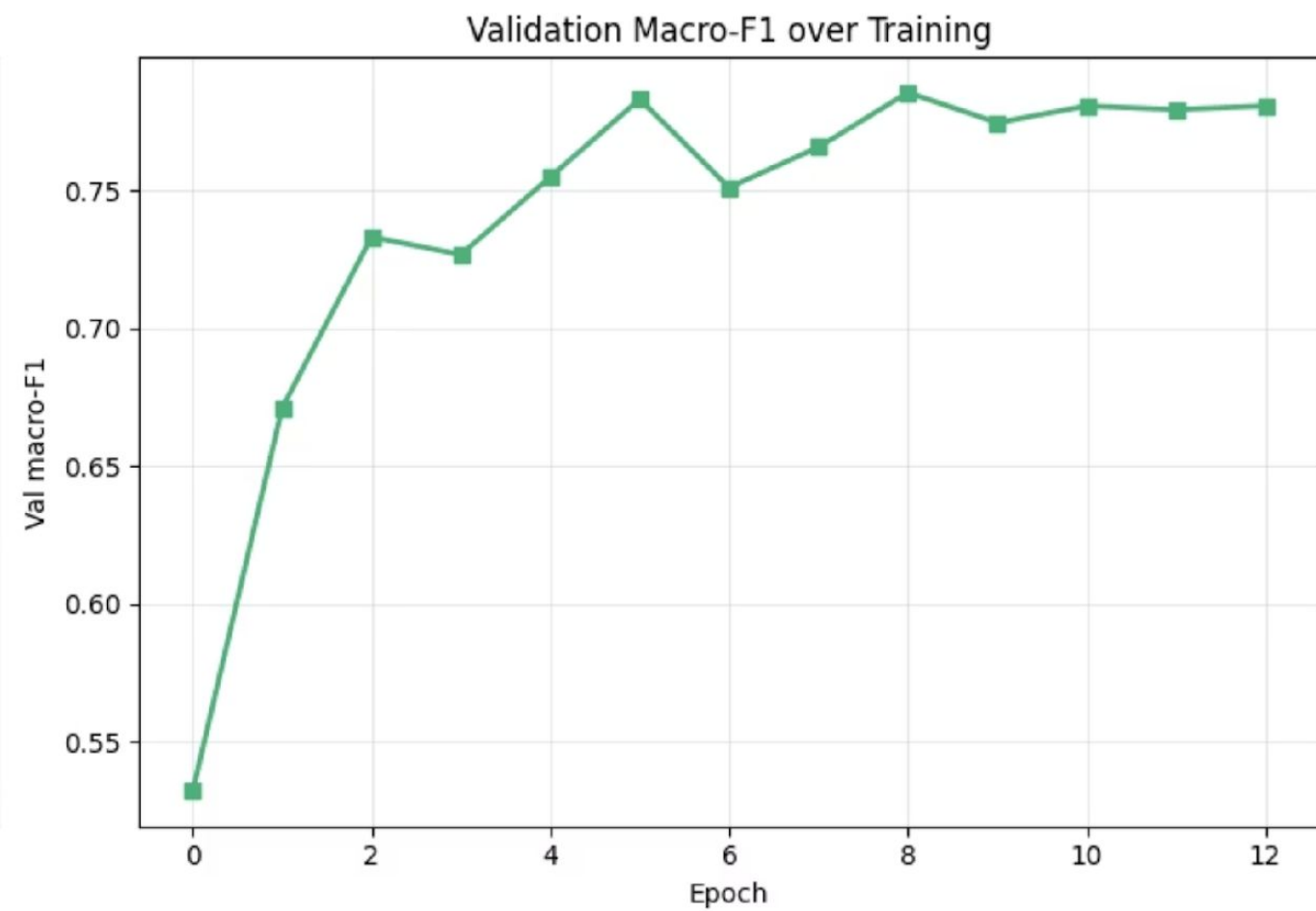
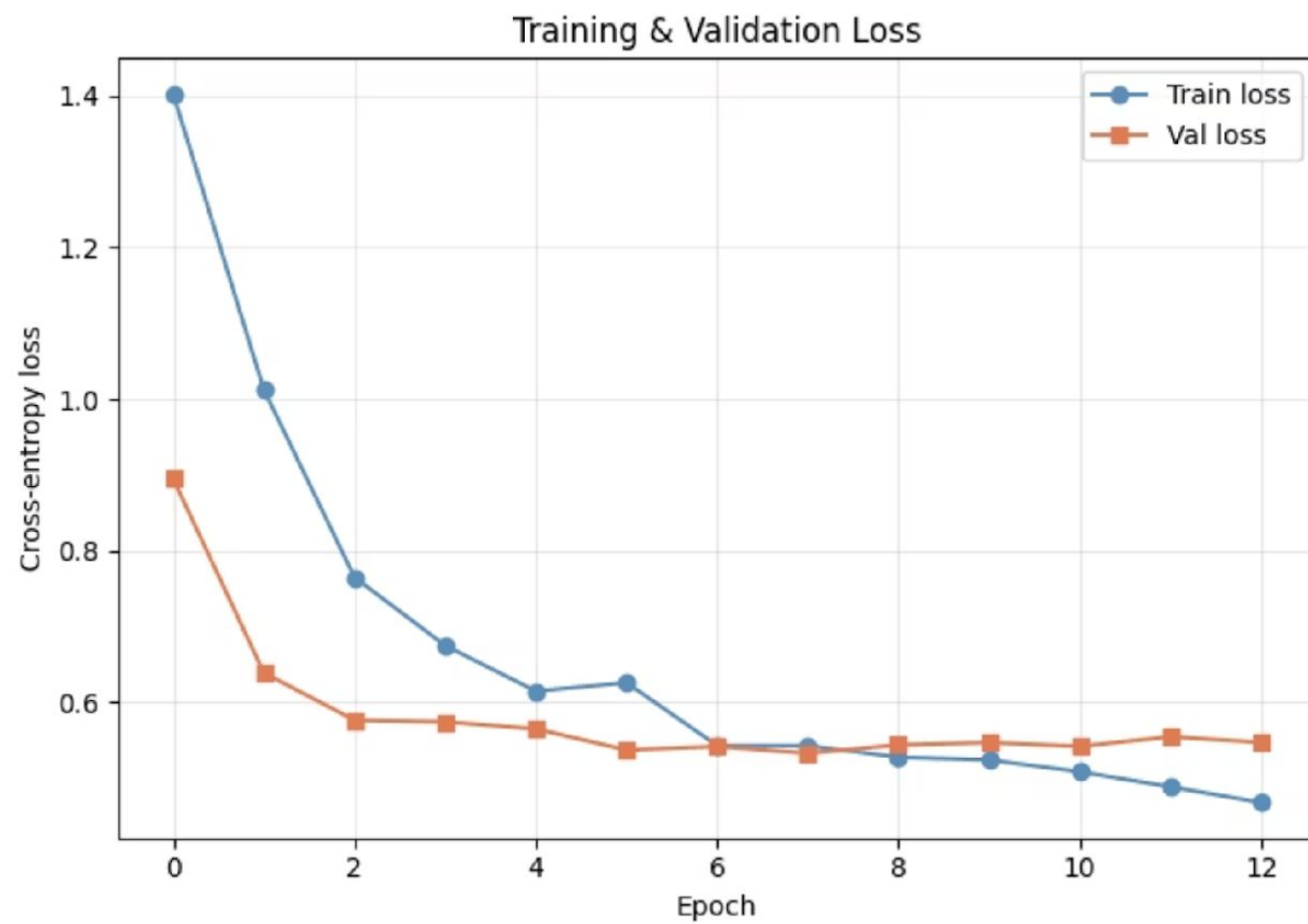
Surprisingly competitive on N $\approx$ 1,000

Six-Model Ablation Table

Configuration	Macro-F1
GMM (probabilistic baseline)	0.330
BERT-only (text stream only)	0.167
Tabular MLP only	0.733
XGBoost (full features)	0.749
SVM RBF (full features)	0.781
DL Full Hybrid (BERT + Tabular)	0.783



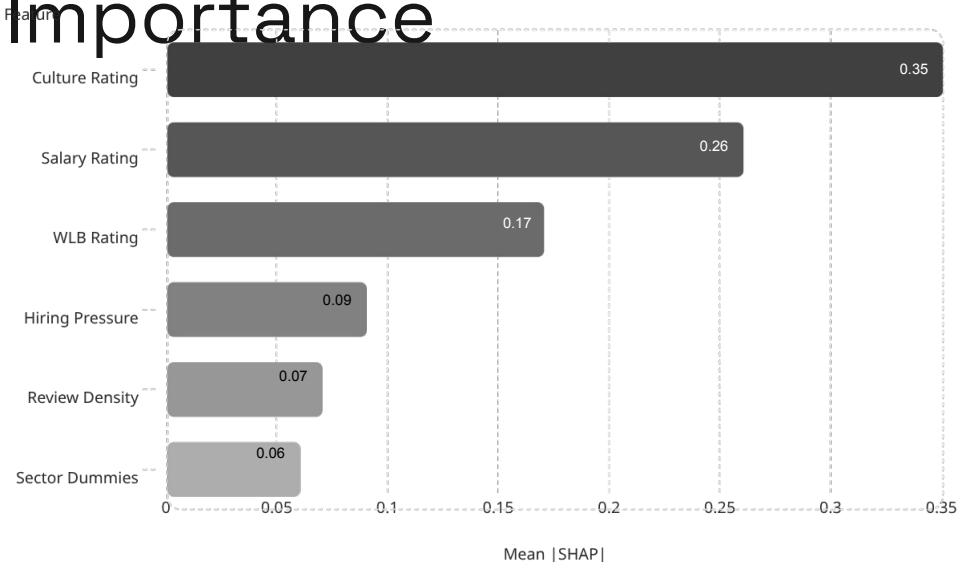
Key insight: BERT alone scores 0.167 — near-useless. Top hiring roles are 2–4 generic titles, identical across classes. Yet fusion *adds* +0.050 F1 over tabular-only (0.783 vs 0.733) — the projection head learns to specialise on the few discriminative role titles (e.g. "AI Engineer", "Semiconductor Engineer").





INTERPRETABILITY

SHAP-Based Feature Importance





Culture Rating — Top Predictor

Mean SHAP: 0.35. The single strongest signal for hiring-quality classification across all models.



Salary & WLB Ratings

Salary (0.26) and work-life balance (0.17) are the second and third strongest predictors — confirming the composite score's feature weighting was well-calibrated.



Engineered Features

Hiring pressure (rank 4th) and review density (rank 5th) — both domain-motivated engineered features — carry meaningful independent signal. Sector dummies appear in the top features, confirming sector matters beyond ratings alone.

✅ CAVITE is **interpretable, not a black box**. SHAP exact Shapley values are computed in polynomial time for tree models, and the feature importance story is coherent with domain knowledge.

Conclusion & Future Directions

1

Fix Leakage

Identified and systematically fixed target leakage that made the original task scientifically invalid (DT F1: 0.985 → 0.65)

2

Benchmark

Compared six configurations — GMM, SVM, XGBoost, BERT-only, Tabular MLP, and Full Hybrid — on the same held-out test set

3

Best Result

DL Hybrid (DistilBERT + Tabular MLP fusion) achieves Macro-F1 = 0.783, marginally topping SVM (0.781) and XGBoost (0.749)

4

Phase 3

Feed GMM soft posteriors as explicit uncertainty features into the fusion layer — closing the loop between probabilistic and neural components

Key Takeaways

Text stream alone is near-useless (F1 = 0.167) for this domain — role titles are too short and too generic — but **fusion still gains +0.050 F1** over tabular-only, confirming the value of multimodal architectures even with weak text

RBF-SVM (0.781) nearly matches the full DL Hybrid (0.783) on $N \approx 1,000$ — a consistent finding in the literature for well-scaled, moderately-sized tabular datasets

SHAP analysis confirms the model is interpretable: **culture, salary, and WLB ratings** are the dominant predictors, consistent with the composite viability score's domain-motivated weights

Thank You

Questions & Discussion

Team CAVITE